SIMATS SCHOOL OF ENGINEERING

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Course Code: DSA03	Course Name: Natural Language Processing
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Assignment 3

1. Draw tree structures for the following ATIS sentences: (10M) (CO3) (BLT4)

a. Does American airlines have a flight between five a.m. and six a.m.
b. I would like to fly on American airlines.
c. Please repeat that.
d. Does American 487 have a first class section?
e. I need to fly between Philadelphia and Atlanta.
f. What is the fare from Atlanta to Denver?
g. Is there an American airlines flight from Philadelphia to Dallas?

2. Consider the below grammar to construct the Early parser for the input string "Christ Opened the eyes". (10M) (CO3) (BLT4)

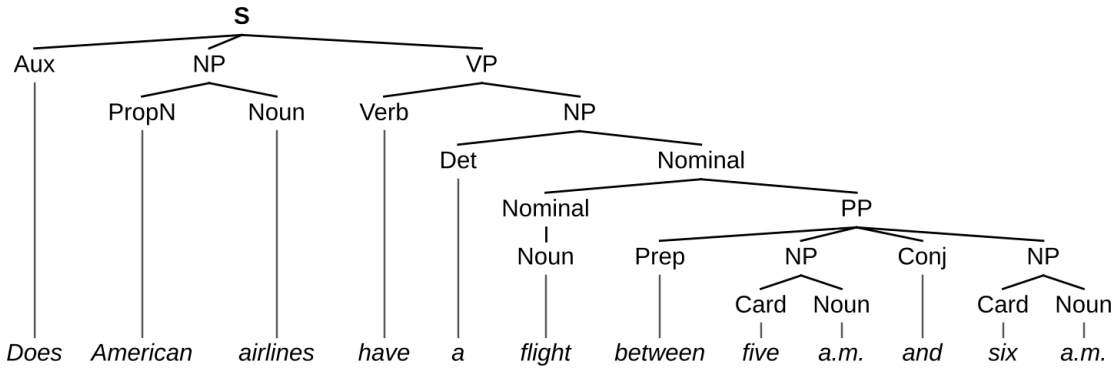
S -> NP VP VP -> V NP PP V -> opened
NP -> N PP -> Prep NP V -> ate
NP -> Det NP N -> Christ Det -> the
NP -> NP NP N -> blind Det -> a
VP -> V N -> eyes Prep -> of

VP -> V NP N -> bread Prep -> in

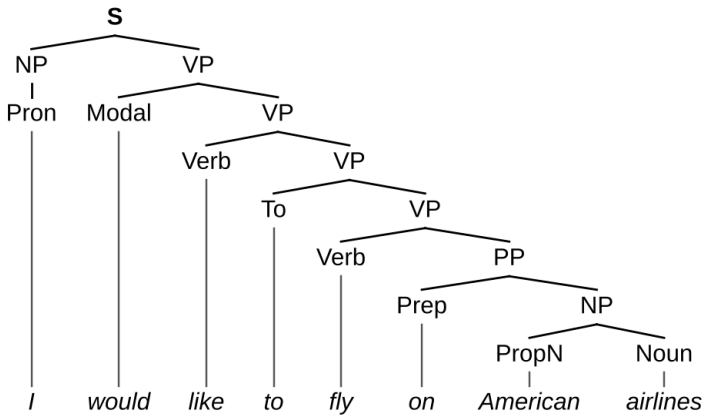
Answer 1. Tree structures for the ATIS sentences

The trees use a small ATIS-style grammar: S → NP VP | Aux NP VP | VP | NP VP | Aux NP NP; NP → Pron | PropN Noun | PropN Card | Det Nominal | NP-conjoined PP; Nominal → Noun | Adj Nominal | Nominal Noun | Nominal PP; VP → Verb NP | Modal VP | To VP | Verb VP | Verb PP | Adv VP; PP → Prep NP (Conj NP). Labels: Aux = auxiliary, PropN = proper noun, Pron = pronoun, Det = determiner, Card = cardinal number, Conj = conjunction, Adv = adverb, WhPron = wh-pronoun, To = infinitive "to". In each tree the sentence words are shown in italics along the bottom.

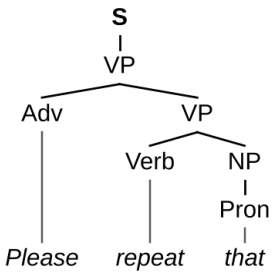
a. Does American airlines have a flight between five a.m. and six a.m.



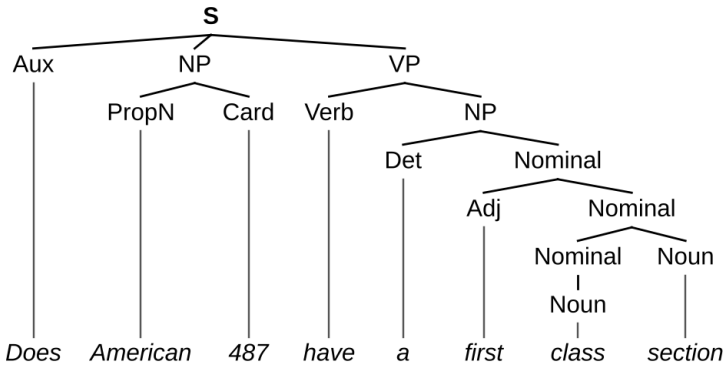
b. I would like to fly on American airlines.



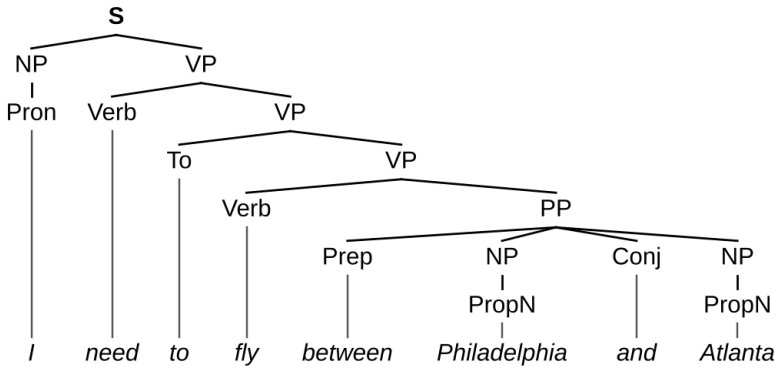
c. Please repeat that.



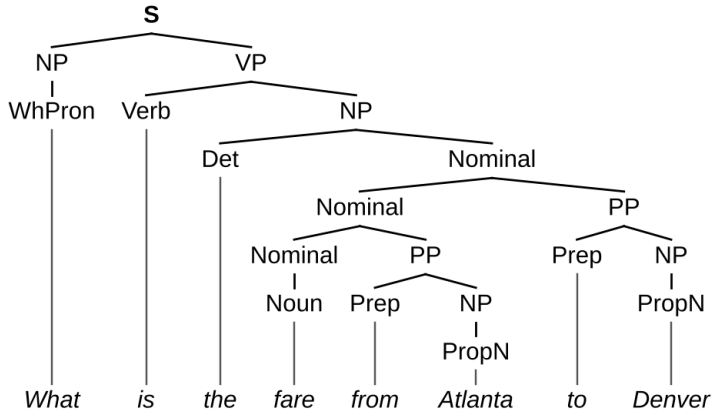
d. Does American 487 have a first class section?



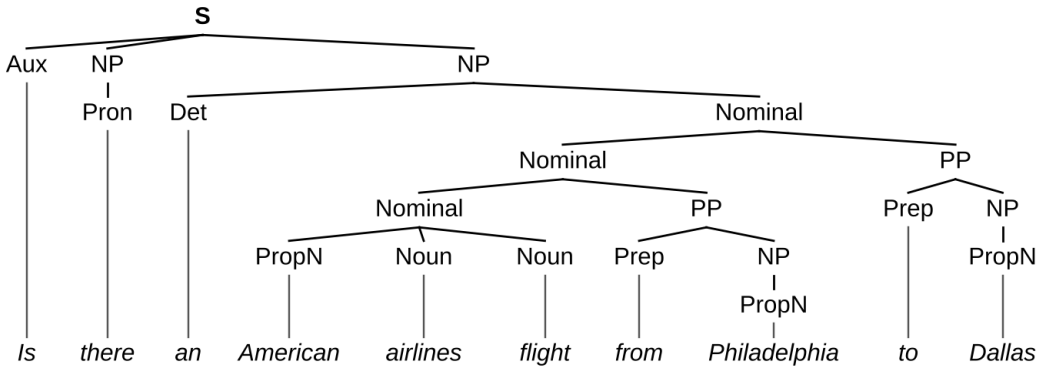
e. I need to fly between Philadelphia and Atlanta.



f. What is the fare from Atlanta to Denver?



g. Is there an American airlines flight from Philadelphia to Dallas?



Answer 2. Earley parse of “Christ opened the eyes”

Grammar read from the question: S → NP VP; NP → N | Det NP | NP NP; VP → V | V NP | V NP PP; PP → Prep NP; V → opened | ate; N → Christ | blind | eyes | bread; Det → the | a; Prep → of | in. The input is normalised to lower case (“Christ opened the eyes”), so positions are: 0 Christ 1 opened 2 the 3 eyes 4. A state is written (rule with • marking progress, [i, j]) where i is where the constituent starts and j is the position of the dot. The start state γ → • S is added to chart S0, and the parse succeeds if γ → S • [0,4] appears in the last chart.

Chart S0

State	Rule	[i, j]	Operation
S0.0	γ → • S	[0, 0]	Start
S0.1	S → • NP VP	[0, 0]	Predictor
S0.2	NP → • N	[0, 0]	Predictor
S0.3	NP → • Det NP	[0, 0]	Predictor
S0.4	NP → • NP NP	[0, 0]	Predictor

Chart S1 (after reading “Christ”)

State	Rule	[i, j]	Operation
S1.0	N → christ •	[0, 1]	Scanner (word “christ”)
S1.1	NP → N •	[0, 1]	Completer
S1.2	S → NP • VP	[0, 1]	Completer
S1.3	NP → NP • NP	[0, 1]	Completer
S1.4	VP → • V	[1, 1]	Predictor
S1.5	VP → • V NP	[1, 1]	Predictor
S1.6	VP → • V NP PP	[1, 1]	Predictor
S1.7	NP → • N	[1, 1]	Predictor
S1.8	NP → • Det NP	[1, 1]	Predictor
S1.9	NP → • NP NP	[1, 1]	Predictor

Chart S2 (after reading “opened”)

State	Rule	[i, j]	Operation
S2.0	V → opened •	[1, 2]	Scanner (word “opened”)
S2.1	VP → V •	[1, 2]	Completer
S2.2	VP → V • NP	[1, 2]	Completer
S2.3	VP → V • NP PP	[1, 2]	Completer
S2.4	S → NP VP •	[0, 2]	Completer
S2.5	NP → • N	[2, 2]	Predictor
S2.6	NP → • Det NP	[2, 2]	Predictor
S2.7	NP → • NP NP	[2, 2]	Predictor
S2.8	γ → S •	[0, 2]	Completer

Chart S3 (after reading “the”)

State	Rule	[i, j]	Operation
S3.0	Det → the •	[2, 3]	Scanner (word “the”)
S3.1	NP → Det • NP	[2, 3]	Completer
S3.2	NP → • N	[3, 3]	Predictor
S3.3	NP → • Det NP	[3, 3]	Predictor
S3.4	NP → • NP NP	[3, 3]	Predictor

Chart S4 (after reading “eyes”)

State	Rule	[i, j]	Operation
S4.0	N → eyes •	[3, 4]	Scanner (word “eyes”)
S4.1	NP → N •	[3, 4]	Completer
S4.2	NP → Det NP •	[2, 4]	Completer
S4.3	NP → NP • NP	[3, 4]	Completer
S4.4	VP → V NP •	[1, 4]	Completer
S4.5	VP → V NP • PP	[1, 4]	Completer
S4.6	NP → NP • NP	[2, 4]	Completer
S4.7	NP → • N	[4, 4]	Predictor
S4.8	NP → • Det NP	[4, 4]	Predictor
S4.9	NP → • NP NP	[4, 4]	Predictor
S4.10	S → NP VP •	[0, 4]	Completer
S4.11	PP → • Prep NP	[4, 4]	Predictor
S4.12	γ → S •	[0, 4]	Completer

Result. State S4.12, γ → S • [0,4], is in the final chart, so the string is accepted. Following the completer links back gives the parse S[0,4] → NP[0,1] VP[1,4], with NP → N (Christ) and VP → V NP, where V = opened and NP[2,4] → Det NP (the + NP → N eyes). The other states in the charts (for example NP → NP • NP) are partial hypotheses created by the left-recursive rule NP → NP NP that never complete, so the sentence has exactly one parse.

